

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (Expected Graduation: May 2027)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Python, Java/Scala, Airflow, Docker, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Developed cloud-hosted **Python/Java** services processing **50M+ daily transactions**, collaborating in **Agile sprints** with code reviews and **CI-gated** releases on enterprise-scale systems.
- Optimized backend performance by migrating to **Spark-native APIs** and tuning resource allocation, reducing **latency by 40%**; shipping **multi-region services** with **fault tolerance** and automated **failover**.

Software Engineer Intern | *React, TypeScript/Node.js, Python/Flask, AWS*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Led **end-to-end feature delivery**: scoped, built, tested, and shipped scheduling UX, multi-brand management, and media publishing with **React** frontends, **Python/Flask** and **Node.js** backends, and **REST APIs** on **AWS**.
- Designed and built **API services** processing **thousands of requests/day**, integrating **OpenAI** for content generation with **NLP** context injection and **API guardrails**.
- Raised reliability to **99.9%** via **Playwright E2E** and integration tests in **GitHub Actions CI/CD**, built **monitoring/alerting** with **CloudWatch**, and hardened auth/session with **retry/backoff** across **Dockerized** stacks.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led **development** of an AI-driven VR clinical simulation in Unreal (C++), collaborating with UBC's School of Nursing to train nurse practitioners.
- Designed a novel **multithreaded** supervisory architecture (C++/**Python**) that monitors **AI** patient behavior and enforces clinical protocol adherence.
- Evaluated system performance, achieving **40%** latency reduction and **90%** inference cost savings through **pipeline optimizations**.

PROJECTS

Backer | *JavaScript/TypeScript, Next.js, AWS (DynamoDB, S3, SES), NextAuth*

Hack The Coast 2026

- Built a full-stack **JavaScript/TypeScript** platform in a **4-person team**; architected a **DynamoDB**-backed data layer (**10+ tables**) with **transactional writes** powering feed, messaging, and investment workflows.
- Designed **12+ API route groups** with **NextAuth** middleware, role-based route guards, and **S3/SES** integrations on **AWS** cloud infrastructure.

Workbase | *TypeScript/Next.js, Prisma/PostgreSQL, AWS Bedrock, Vitest*

Personal Project

- Built a full-stack career content platform using **Next.js App Router**, **Prisma/PostgreSQL**, and **GitHub OAuth** ingestion to extract evidence from repos and generate verified resume content.
- Designed an AI-backed highlight generation and verification pipeline via **AWS Bedrock Converse API** with **Zod** schema validation, structured output repair, and evidence attribution tracing.
- Wrote **Vitest** test suites covering domain logic (status transitions, eligibility, deduplication) and the structured LLM client (parsing, repair, validation).

SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, HTML/CSS, Scala, C/C++

Platforms & Tools: React/Next.js, Node.js/Express, Flask/FastAPI, REST APIs; AWS (Lambda, DynamoDB, S3, SQS); GitHub, Playwright, Docker, Git, RDBMS